

Functional Requirements Specification

Project: Issue and Defect Tracking System in SAP
Document type: Functional Requirements Specification (FRS)
Language: English
Status: Draft v1.5 Last updated: 2026-07-11 Prepared for: SAP490 project delivery, mentor review, implementation evidence, and QA test design Document style: SAP490 hybrid, function-detail-first, aligned with BRD v1.3 and SRS v1.2

1. Document Control

1.1 Version History

Version	Date	Author	Reviewer	Change Summary	Approval Status
v1.0	2026-06-02	IDTS Project Team	Mentor / Supervisor	Initial FRS created from BRD v1.1, SRS v1.0, BA baseline, diagrams, and SAP490 guidance.	Draft
v1.1	2026-06-03	IDTS Project Team	Mentor / Supervisor	Fixed Mermaid syntax in rejected follow-up sequence and added missing workflow diagrams for create/assign, developer review, request information, retest/closure, and PM monitoring.	Draft
v1.2	2026-06-03	IDTS Project Team	Mentor / Supervisor	Updated functional actors and workflows to	Draft

Version	Date	Author	Reviewer	Change Summary	Approval Status
				the MVP role baseline: Tester, Developer, and PM. Reporter and Admin are deferred as separate roles.	
v1.3	2026-07-10	IDTS Project Team	Mentor / Supervisor	Synced real draft attachment behavior and optional advisory-AI review behavior with the implemented CAP/Fiori baseline.	Draft
v1.4	2026-07-11	IDTS Project Team	Mentor / Supervisor	Replaced eight review-facing Mermaid blocks with rendered workflow figures and closed the visual-submission open issue.	Draft
v1.5	2026-07-11	IDTS Project Team	Mentor / Supervisor	Replaced review figures with editable draw.io sources and readable PNG/PDF review	Draft

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outputs.					

1.2 Review and Sign-Off

Role	Name	Responsibility	Status	Date
Prepared by	IDTS Project Team	Prepare and maintain FRS	Drafted	2026-06-02
Reviewed by	Mentor / Supervisor	Review functional completeness and SAP490 fit	Pending	TBD
Approved by	Mentor / Supervisor	Approve FRS for implementation and test case design	Pending	TBD
Project owner	Team / PM	Confirm MVP functional priority	Pending	TBD

2. Purpose and Scope

This FRS specifies detailed functional behavior for the IDTS MVP. It describes how users and the system shall perform bug creation, duplicate checking, classification, assignment, developer review, request for information, rejection follow-up, status transitions, comments, history logs, notification records, and PM monitoring.

The FRS is more detailed than the BRD and more workflow-oriented than the SRS. It is intended for CAP/Fiori implementation planning and QA test case design. It does not contain source code, final CDS schema, Fiori annotation code, UI5 controller code, or deployment credentials.

3. Functional Scope Summary

Functional Area	Included in MVP	Notes
Bug creation	Yes	Structured report with required fields and unique bug number.
Duplicate checking	Yes	Manual search/filter remains the core flow; advisory AI similar-bug suggestion is optional and review-only.
Classification	Yes	SAP Module optional; Application Component and Defect Category required.

Functional Area	Included in MVP	Notes
Developer matching	Yes	Based on Component Category and optional SAP Module.
Assignment	Yes	Assigned or Pending Assignment.
Developer review	Yes	Start review, request information, reject, progress, resolve.
Rejected follow-up	Yes	Rejected is not final; reason and nextProcessor required.
Retest and closure	Yes	Retest Required before close when verification is needed.
Comments	Yes	Discussion attached to bug; no direct status change.
History log	Yes	Important actions recorded.
Notification records	Yes	Event records and triggers; external delivery can be deferred.
PM monitoring	Yes	Workload, overdue, queues, nextProcessor, status filters.
Attachments	Yes	Draft upload/download and evidence metadata are implemented; attachment bytes are never sent to AI.
Advisory AI	Optional	Similar-bug, classification, summary, and Smart Assign explanations require human review and never mutate the bug.

4. Functional Workflow Diagrams

4.1 Main Defect Tracking Flow

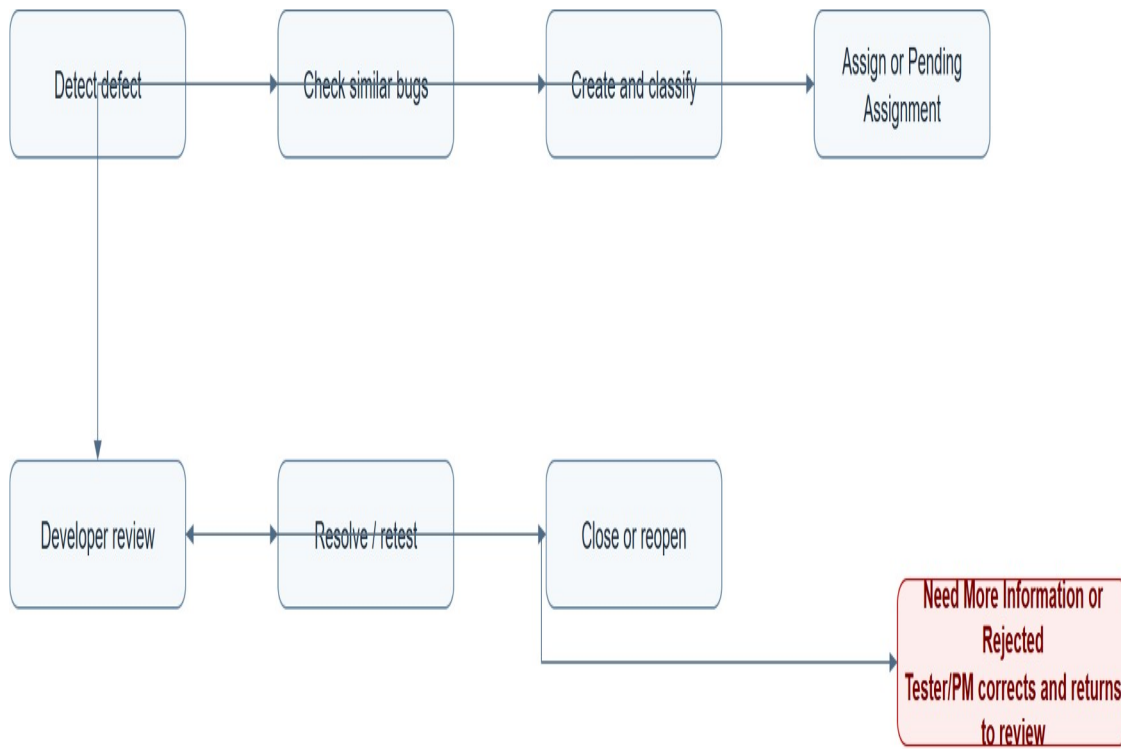


Figure 14. Main Defect Tracking Flow. IDTS Diagram Pack; editable master is maintained as a draw.io file with the project artifacts.

4.2 Rejected Follow-up Flow

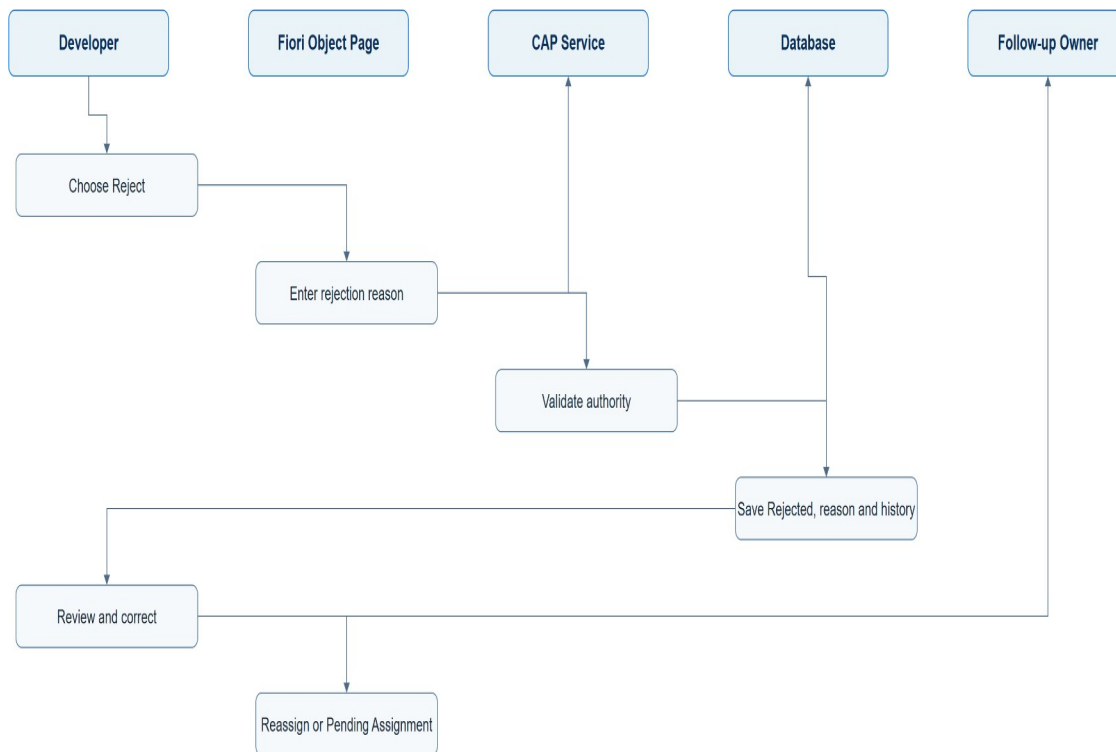


Figure 15. Rejected Follow-up Flow. IDTS Diagram Pack; editable master is maintained as a draw.io file with the project artifacts.

4.3 Status Lifecycle

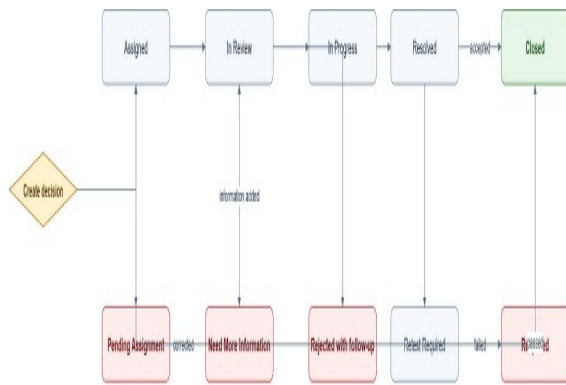


Figure 16. Status Lifecycle. IDTS Diagram Pack; editable master is maintained as a draw.io file with the project artifacts.

4.4 Bug Creation and Assignment Activity Flow

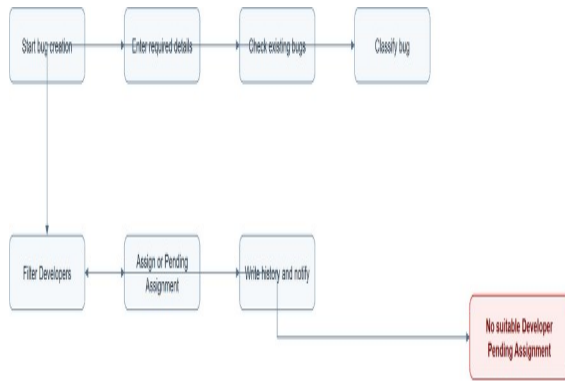


Figure 17. Bug Creation and Assignment Activity Flow. IDTS Diagram Pack; editable master is maintained as a draw.io file with the project artifacts.

4.5 Developer Review Decision Flow

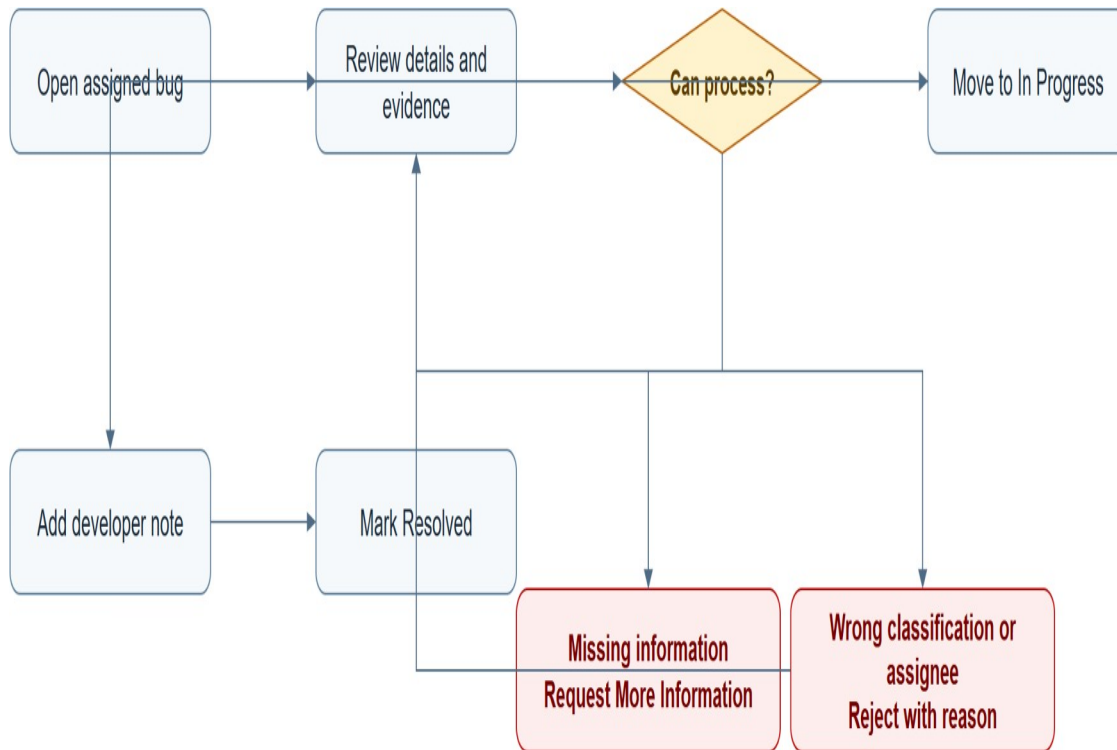


Figure 18. Developer Review Decision Flow. IDTS Diagram Pack; editable master is maintained as a draw.io file with the project artifacts.

4.6 Request More Information Flow

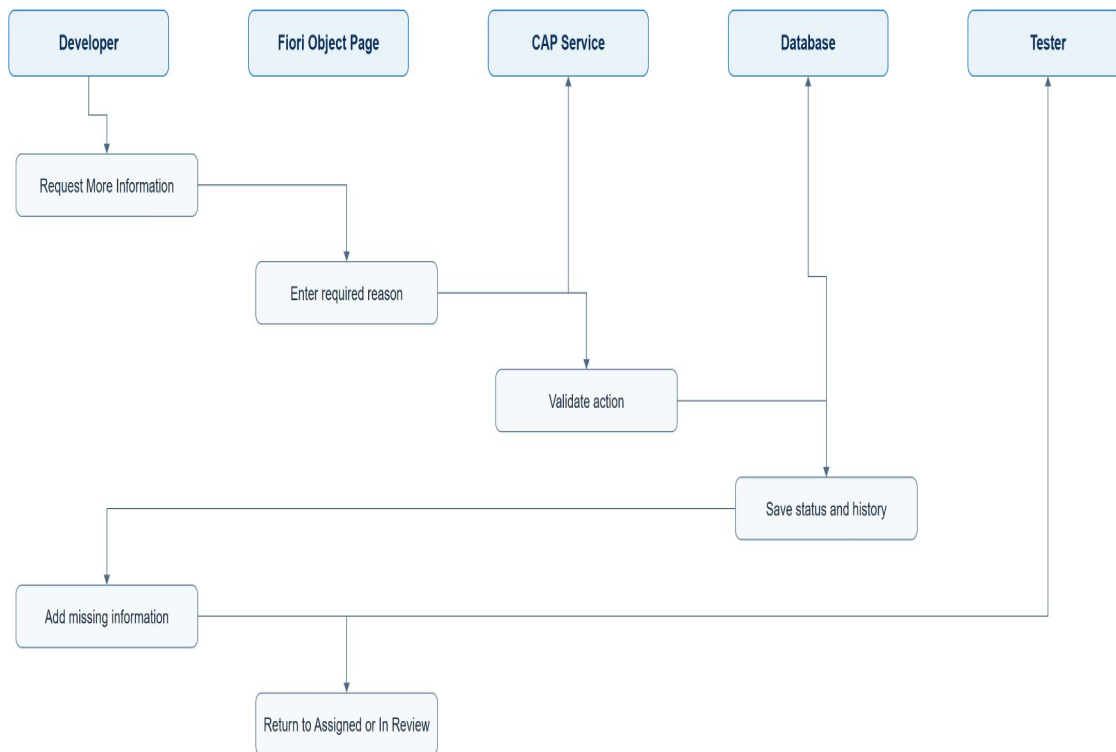


Figure 19. Request More Information Flow. IDTS Diagram Pack; editable master is maintained as a draw.io file with the project artifacts.

4.7 Resolve, Retest, Close, and Reopen Flow

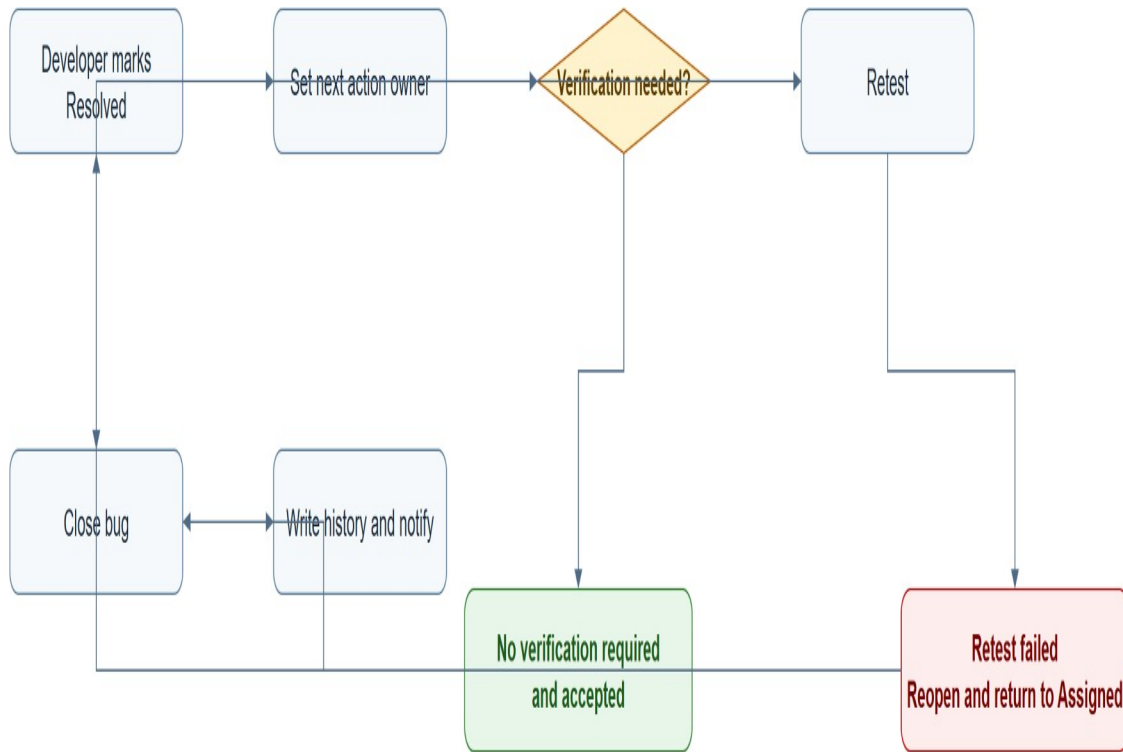


Figure 20. Resolve, Retest, Close, and Reopen Flow. IDTS Diagram Pack; editable master is maintained as a draw.io file with the project artifacts.

4.8 PM Monitoring and Escalation Flow

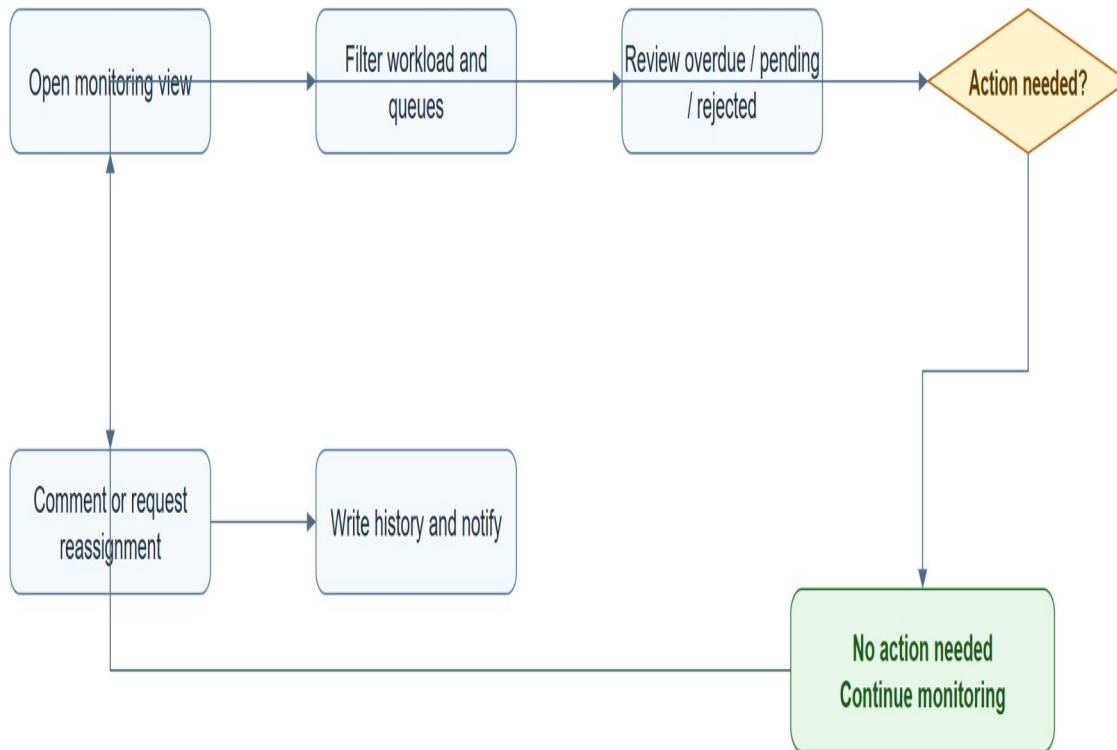


Figure 21. PM Monitoring and Escalation Flow. IDTS Diagram Pack; editable master is maintained as a draw.io file with the project artifacts.

5. Detailed Functional Requirements

5.1 FRS-BUG-001 - Create Bug Report

Field	Specification
Purpose	Capture a structured defect report so the team can review, assign, process, and verify it.
Primary actor	Tester.
Trigger	User chooses to create a new bug after checking existing bugs or deciding no existing bug is suitable.
Preconditions	User is authorized to create bugs; required value-help data is available; duplicate search has been performed or intentionally skipped with user decision.
Main flow	User enters title, description, priority, severity, environment, steps to reproduce, actual result, expected result, optional SAP Module, Application Component, Defect Category, optional testCaseRef, optional testRunRef, and optional evidence metadata;

Field	Specification
	system validates required fields; user selects Developer or No suitable developer; system submits the bug.
Alternative flow	If the user finds a similar open bug, the user follows, comments, or updates that existing bug instead of creating a new record.
Validation rules	Title, description, priority, severity, Application Component, Defect Category, steps to reproduce, actual result, expected result, and assignment decision are required for submit. SAP Module is optional.
Status effect	Selected Developer creates status Assigned. No suitable Developer creates status Pending Assignment.
History effect	Create and assignment decision must be logged.
Notification effect	Assigned bug creates a Developer notification record. Pending Assignment creates a PM or Tester queue notification record.
Acceptance criteria	Bug is created with unique bugNumber; required fields block invalid submit; Assigned and Pending Assignment outcomes work; history log exists.
Traceability	SRS-FR-BUG-001, SRS-FR-BUG-002, SRS-DATA-001, SRS-DATA-002.

5.2 FRS-BUG-002 - Check Existing Bugs and Duplicate Support

Field	Specification
Purpose	Reduce duplicate reports and guide users to existing bug records when appropriate.
Primary actor	Tester.
Trigger	User starts bug creation or searches bug list.
Preconditions	Bug list is accessible to the user according to authorization.
Main flow	User searches by title, keyword, status, priority, severity, SAP Module, Application Component, Defect Category, assignee, reporter, created date, or updated date; user reviews similar results; user decides whether to create new, follow existing open bug, or reopen/update closed bug.

Field	Specification
Alternative flow	If no similar bug is found, user continues bug creation.
Validation rules	Reopen closed bug requires reason and authorized role.
Status effect	Existing open bug keeps its status unless a separate authorized action changes it. Closed bug can move to Reopened if allowed.
History effect	Reopen or duplicate-link action must be logged.
Acceptance criteria	Search/filter works; user can avoid new duplicate; closed bug reopen path is controlled; duplicate link can be added when implemented.
Traceability	SRS-FR-BUG-003, SRS-FR-BUG-004.

5.3 FRS-CLASS-001 - Classify Bug

Field	Specification
Purpose	Classify the bug clearly for assignment and PM reporting.
Primary actor	Tester.
Trigger	User creates or edits a non-closed bug.
Preconditions	Classification master data is active and available.
Main flow	User selects SAP Module if the bug belongs to a SAP business context; user selects Application Component; user selects Defect Category; system resolves Component Category.
Alternative flow	For pure IDTS bugs, SAP Module should remain empty.
Validation rules	Application Component and Defect Category are required; Component Category must be an active valid pair; invalid pairs are rejected.
UI behavior	Application Component value help can be filtered by selected SAP Module; Defect Category value help can be filtered by Application Component.
Status effect	Classification changes do not automatically close or resolve a bug.
History effect	Important classification changes must be logged, especially after Rejected follow-up.

Field	Specification
Acceptance criteria	SAP Module, Application Component, and Defect Category are separate; invalid component/category pair is not accepted; Developer candidate list reflects selected classification.
Traceability	SRS-FR-CLASS-001 to SRS-FR-CLASS-005.

5.4 FRS-ASSIGN-001 - Filter and Assign Developer

Field	Specification
Purpose	Assign a bug to a suitable Developer using responsibility mapping.
Primary actor	Tester; PM if direct assignment is approved.
Trigger	User submits a bug or chooses Assign/Reassign.
Preconditions	Bug has valid Application Component and Defect Category; Component Category can be resolved.
Main flow	System uses Component Category and optional SAP Module to filter active Developer Responsibilities; UI shows suitable Developer candidates; user selects one Developer; system assigns bug.
Alternative flow	If no Developer matches or all candidates are unsuitable, user selects No suitable developer.
Validation rules	Selected Developer must have an active profile and active responsibility for the Component Category and optional SAP Module scope; one bug has one main assignee at a time.
Status effect	Assignment from New or Pending Assignment sets status Assigned. Reassignment keeps status meaningful to the current flow unless a specific allowed transition is chosen.
History effect	Assign and reassign actions must log old assignee, new assignee, actor, timestamp, and reason when applicable.
Notification effect	Assigned or reassigned Developer gets a notification record.
Acceptance criteria	Candidate list is filtered; invalid assignee is rejected by backend; assignment creates

Field	Specification
Traceability	history and notification records. SRS-FR-ASSIGN-001, SRS-FR-ASSIGN-002.

5.5 FRS-ASSIGN-002 - Submit as Pending Assignment

Field	Specification
Purpose	Keep valid bugs visible when no suitable Developer is available.
Primary actor	Tester.
Trigger	User submits a valid bug with No suitable developer.
Preconditions	Required bug fields and classification are valid.
Main flow	User chooses No suitable developer; system submits bug; system sets status Pending Assignment; system sets nextProcessor to PM queue or Tester; system creates history and notification records.
Alternative flow	PM or Tester later assigns a suitable Developer and moves bug to Assigned.
Validation rules	Pending Assignment is allowed only for valid bug reports, not as a way to bypass required fields.
Status effect	New to Pending Assignment. Pending Assignment to Assigned when a suitable Developer is selected.
History effect	Pending Assignment entry and later assignment must be logged.
Acceptance criteria	Bug is not lost; Pending Assignment appears in monitoring; follow-up assignment works.
Traceability	SRS-FR-ASSIGN-003, SRS-FR-PM-003.

5.6 FRS-ASSIGN-003 - Reassign Bug

Field	Specification
Purpose	Move responsibility to another Developer when the current assignment is unsuitable or capacity changes.
Primary actor	Tester; PM if direct reassignment is approved.
Trigger	User chooses Reassign or handles rejected follow-up.

Field	Specification
Preconditions	Bug is not Closed or user has reopen authority; new Developer passes responsibility validation.
Main flow	User enters reason when required; user selects new Developer from filtered candidates; system updates assignee, nextProcessor, history, and notification records.
Alternative flow	If no suitable Developer exists, user moves bug to Pending Assignment.
Validation rules	Reassign is an action/history event, not a primary status. Reason is required for rejected follow-up, wrong assignment, or PM escalation.
Status effect	Status can remain Assigned/In Review/In Progress when only assignee changes; Rejected can move to Assigned or Pending Assignment after correction.
Acceptance criteria	Old and new assignee are visible in history; new assignee is valid; Rejected follow-up cannot be left without owner.
Traceability	SRS-FR-ASSIGN-004, SRS-FR-STATUS-004.

5.7 FRS-DEV-001 - Developer Review, Progress, and Resolve

Field	Specification
Purpose	Let Developer review and process assigned bugs without treating IDTS as a code fixing tool.
Primary actor	Developer.
Trigger	Developer opens My Assigned Bugs or receives notification.
Preconditions	Bug is assigned to Developer or Developer is otherwise authorized.
Main flow	Developer opens bug details; reviews classification, reproduction steps, evidence, comments, and history; starts review; adds note or comment; moves valid bug to In Progress; marks Resolved when response or resolution result is complete.
Alternative flow	Developer requests more information or rejects unsuitable classification/assignment.

Field	Specification
Validation rules	Developer cannot close bug directly in recommended MVP; Developer can only update assigned or authorized bugs.
Status effect	Assigned to In Review; In Review to In Progress; In Progress to Resolved.
History effect	Start review, progress, developer note, and resolved actions should be logged where meaningful.
Notification effect	Status update or resolved event creates Tester/PM notification record when applicable.
Acceptance criteria	Developer can perform review flow; direct close is not available for Developer; history records status changes.
Traceability	SRS-FR-STATUS-001, SRS-FR-STATUS-005.

5.8 FRS-INFO-001 - Request More Information

Field	Specification
Purpose	Prevent unclear bugs from being processed without enough information.
Primary actor	Developer.
Trigger	Developer identifies missing or unclear information.
Preconditions	Bug is Assigned or In Review and Developer is assigned or authorized.
Main flow	Developer selects Request More Information; system requires reason; system sets status Need More Information; system sets nextProcessor to Tester; system writes history and notification record.
Alternative flow	Tester updates missing information and returns bug to Assigned or In Review.
Validation rules	Reason is required; closed bugs cannot enter Need More Information without Reopen.
Status effect	Assigned or In Review to Need More Information; then Need More Information to Assigned or In Review after Tester action.
Acceptance criteria	Reason is required; Tester can see required action; Developer can continue review after information is supplied.

Field	Specification
Traceability	SRS-FR-STATUS-002, SRS-FR-STATUS-009.

5.9 FRS-REJECT-001 - Reject Unsuitable Assignment or Classification

Field	Specification
Purpose	Give Developer a controlled rejection path without making Rejected a final state.
Primary actor	Developer, then Tester or PM as follow-up owner.
Trigger	Developer determines the bug is wrongly classified, wrongly assigned, or outside responsibility.
Preconditions	Bug is Assigned or In Review; Developer is assigned or authorized.
Main flow	Developer selects Reject; system requires rejection reason; system sets status Rejected; system sets nextProcessor to Tester or PM; system writes history and notification record; follow-up owner reviews reason, corrects classification or information, then reassigns to a suitable Developer or moves bug to Pending Assignment.
Alternative flow	PM comments or requests reassignment if PM direct reassignment is not approved.
Validation rules	Rejection reason is required; nextProcessor is required unless represented by a configured role queue; Rejected cannot transition directly to Closed; follow-up transition must be Assigned or Pending Assignment.
Status effect	Assigned/In Review to Rejected; Rejected to Assigned or Pending Assignment after correction.
History effect	Reject reason, actor, old status, new status, and follow-up action must be logged.
Notification effect	Tester and PM when applicable receive notification records.
Acceptance criteria	Rejected bug is visible as a follow-up queue; reason is shown on Object Page; no rejected bug is left without owner/action.
Traceability	SRS-FR-STATUS-003, SRS-FR-STATUS-004, SRS-FR-STATUS-009.

5.10 FRS-STATUS-002 - Resolve, Retest, and Close

Field	Specification
Purpose	Prevent bug closure before verification when retest is needed.
Primary actor	Developer, Tester/PM.
Trigger	Developer marks a bug Resolved.
Preconditions	Bug is In Progress and Developer is assigned or authorized.
Main flow	Developer marks Resolved with note when needed; system sets nextProcessor to Tester/PM; Tester/PM sends bug to Retest Required if verification is needed; if retest passes, Tester/PM closes the bug.
Alternative flow	If no retest is needed and result is accepted, Tester/PM can close directly from Resolved when rule allows.
Validation rules	Developer should not close directly in recommended MVP. Close requires authorized Tester/PM.
Status effect	In Progress to Resolved; Resolved to Retest Required or Closed; Retest Required to Closed.
History effect	Resolve, retest, and close actions must be logged.
Acceptance criteria	Resolved does not silently become Closed; retest pass closes; action history is visible.
Traceability	SRS-FR-STATUS-006, SRS-FR-STATUS-007.

5.11 FRS-STATUS-003 - Reopen Bug

Field	Specification
Purpose	Continue processing when a resolved or closed bug still exists.
Primary actor	Tester/PM.
Trigger	Retest fails or user discovers the issue still exists after closure.
Preconditions	User is authorized to reopen.
Main flow	User selects Reopen; system requires reason; system sets status Reopened; user updates information or assigns Developer; system logs history and notification.

Field	Specification
Alternative flow	If the bug is still under Retest Required, retest failure moves to Reopened.
Validation rules	Reopen reason is required; Closed bugs should not be freely edited without Reopen.
Status effect	Resolved, Retest Required, or Closed to Reopened; Reopened to Assigned after reassignment.
Acceptance criteria	Reason is captured; processing can continue; history preserves closure and reopen path.
Traceability	SRS-FR-BUG-005, SRS-FR-STATUS-007.

5.12 FRS-STATUS-005 - Status Transition Validation

Field	Specification
Purpose	Ensure lifecycle consistency across UI and backend.
Primary actor	System.
Trigger	Any status-changing action or update.
Preconditions	Current bug status is known.
Main flow	System checks requested transition against the status transition matrix; system validates actor role and required reason; system rejects invalid transition with actionable error message.
Alternative flow	PM escalation override can be considered only if explicitly approved later.
Validation rules	Backend validation is authoritative; hidden Fiori actions do not replace backend checks.
Status effect	Only allowed transitions persist.
Acceptance criteria	Invalid transitions fail; required reason is enforced; accepted transition logs history.
Traceability	SRS-FR-STATUS-008, SRS-NFR-INT-001.

5.13 FRS-NEXTP-001 - Maintain nextProcessor

Field	Specification
Purpose	Show who or which queue must act next.
Primary actor	System.
Trigger	Bug creation, assignment, reassignment, status change, request information, reject, resolve, retest, close, or reopen.
Preconditions	Action and target status are known.

Field	Specification
Main flow	System sets nextProcessor using mapping rules: Assigned/In Review/In Progress to assigned Developer; Need More Information to Tester; Pending Assignment to PM queue or Tester; Rejected to Tester or PM; Resolved/Retest Required to Tester/PM; Closed to empty.
Alternative flow	PM may override nextProcessor only for escalation or exceptional reassignment if approved.
Validation rules	nextProcessor does not replace assignee. Rejected must have nextProcessor or a configured follow-up queue.
History effect	Important nextProcessor changes should be logged.
Acceptance criteria	My Action Items and PM queues reflect nextProcessor; Closed has no nextProcessor; Rejected has a clear follow-up owner.
Traceability	SRS-FR-STATUS-009.

5.14 FRS-COMMENT-001 - Add Comments

Field	Specification
Purpose	Keep collaboration attached to a bug record.
Primary actor	Tester, Developer, PM.
Trigger	Authorized user adds a comment on Object Page.
Preconditions	User can view the bug and is allowed to comment.
Main flow	User enters comment content; system stores content, author, author role, timestamp, and bug reference; system displays comment list in chronological order.
Alternative flow	Comment can support a later status action, but status must be changed through a separate authorized action.
Validation rules	Comment content is required; comment cannot directly change status.
History effect	Comment event may be logged when audit policy requires.
Acceptance criteria	Comment is visible on bug; status remains unchanged; author and timestamp are stored.

Field	Specification
Traceability	SRS-FR-COMMENT-001, SRS-FR-COMMENT-002.

5.15 FRS-AUDIT-001 - History Log

Field	Specification
Purpose	Preserve traceability of important changes.
Primary actor	System.
Trigger	Important business action occurs.
Preconditions	Bug and actor context are available.
Main flow	System records bug reference, actor, actor role, timestamp, action type, old value, new value, and reason when applicable.
Logged actions	Create, edit, assign, reassign, status change, request information, reject, resolve, retest, close, reopen, comment event, attachment event, notification event.
Validation rules	Reason is required for reject, reopen, request information, and selected reassignment cases.
Acceptance criteria	History answers who did what, when, and why where applicable.
Traceability	SRS-FR-AUDIT-001, SRS-FR-AUDIT-002.

5.16 FRS-NOTIF-001 - Notification Records

Field	Specification
Purpose	Record important notification events without hardcoding delivery channels.
Primary actor	System.
Trigger	Assignment, reassignment, request information, bug update, rejection, overdue, resolved, retest, or close event.
Preconditions	Event and recipient can be determined.
Main flow	System creates notification record with bug reference, recipient, eventType, channel where known, deliveryStatus, and timestamp.
Alternative flow	External delivery adapter can be added later; failure to deliver shall not remove history log.
Validation rules	No private endpoint, webhook URL, token, or service key is stored in repo code.
Acceptance criteria	Notification records exist for important events; Rejected notification makes follow-up

Field	Specification
	owner clear.
Traceability	SRS-FR-NOTIF-001, SRS-FR-NOTIF-002.

5.17 FRS-PM-001 - PM Monitoring Dashboard and Lists

Field	Specification
Purpose	Help PM monitor risk, workload, overdue items, and ownership.
Primary actor	PM.
Trigger	PM opens List Report, dashboard, or monitoring view.
Preconditions	PM has monitoring authorization.
Main flow	PM filters bugs by status, priority, severity, SAP Module, Application Component, Defect Category, assignee, nextProcessor, created date, updated date, due date, and overdue state; PM reviews workload by Developer and queues.
Queue views	All Bugs, My Action Items, Pending Assignment, Need More Information, Retest Required, Rejected Follow-up, Overdue, My Assigned Bugs where relevant.
Validation rules	PM can view all bugs; PM direct reassignment depends on authorization decision.
Acceptance criteria	PM can identify unassigned, overdue, rejected, stale, and overloaded areas quickly.
Traceability	SRS-FR-PM-001, SRS-FR-PM-002, SRS-FR-PM-003.

5.18 FRS-PM-002 - PM Reassignment Request or Direct Reassignment

Field	Specification
Purpose	Let PM coordinate reassignment without replacing Developer or Tester responsibilities.
Primary actor	PM.
Trigger	PM detects overload, overdue, wrong assignment, repeated rejection, or long pending item.
Preconditions	PM can view the bug.
Main flow	PM comments or requests reassignment; Tester performs reassignment by default.

Field	Specification
Alternative flow	If mentor/team approves PM direct reassignment, PM selects a valid Developer and system logs the action.
Validation rules	PM direct reassignment must be explicitly enabled in authorization; reason should be recorded.
Acceptance criteria	PM can coordinate reassignment; ownership stays clear; action is logged.
Traceability	SRS-FR-PM-004.

5.19 FRS-UX-001 - Fiori List Report and Object Page Behavior

Field	Specification
Purpose	Provide a usable enterprise UI aligned with Fiori Elements.
Primary actor	Tester, Developer, PM.
Trigger	User opens bug management app.
Preconditions	OData service and annotations are available.
List Report behavior	Show bugNumber, title, status, priority, severity, SAP Module, Application Component, Defect Category, assignee, nextProcessor, due date, updatedAt, and createdAt where available.
Filter behavior	Support status, priority, severity, SAP Module, Application Component, Defect Category, assignee, nextProcessor, overdue, created date, and updated date filters.
Object Page sections	Bug Details, Classification, Assignment, Reproduction, Comments, Attachments, History, Notifications, PM Monitoring.
Message behavior	Use inline value states for missing fields, Message Popover for multiple issues, and confirmation dialogs for Reject, Reopen, Close, or destructive attachment removal.
Semantic colors	Positive for Resolved and Closed; Critical for Pending Assignment, Need More Information, Retest Required, Overdue; Negative for Rejected; Neutral or Information for New, Assigned, In Review, In Progress, Reopened.
Acceptance criteria	Core fields are visible; dependent value helps are understandable; actions match role/status;

Field	Specification
Traceability	Rejected details are visible; validation messages are actionable. SRS-IF-UI-001 to SRS-IF-UI-005, SRS-NFR-USE-001, SRS-NFR-USE-002.

5.20 FRS-AI-001 - Optional Advisory AI Review

Field	Specification
Purpose	Help users inspect similar bugs, possible classification, concise bug/handoff context, and Smart Assign rationale without automating a business decision.
Primary actor	Tester, Developer, PM.
Trigger	An authorized user explicitly requests an AI suggestion from its relevant business context.
Preconditions	The feature is enabled with approved private configuration, or the UI presents the safe unavailable state while the normal workflow remains usable.
Main flow	CAP allowlists and sanitizes permitted bug/context fields; the provider returns normalized advisory output; the UI renders it as text requiring review; the user may ignore it or use normal actions to make a decision.
Validation rules	AI cannot bypass role authorization, catalog validation, Developer Responsibility eligibility, required reasons, or status transition rules. It does not create, edit, assign, reclassify, or transition bugs.
Data protection	Do not send credentials, tokens, private endpoints, attachment bytes, raw logs, or unnecessary personal data. Do not persist raw prompts/provider responses or hidden reasoning.
Acceptance criteria	Disabled, timeout, malformed, unsafe, and provider-failure states are safe; suggestions are visibly review-only; an audit record is sanitized; normal non-AI workflow continues.
Traceability	SRS-FR-AI-001 to SRS-FR-AI-003; docs/ba/discovery/idts-63-ai-assistance-guardrails.md.

6. Functional Data Rules

Rule ID	Rule
FRS-DATA-RULE-001	Application Component and Defect Category are required for submit.
FRS-DATA-RULE-002	SAP Module is optional and must not be confused with IDTS Application Component.
FRS-DATA-RULE-003	Component Category must be a valid active pair.
FRS-DATA-RULE-004	Developer candidate must have active Developer Responsibility.
FRS-DATA-RULE-005	A bug has one main assignee at a time.
FRS-DATA-RULE-006	Rejected requires reason and nextProcessor or configured follow-up queue.
FRS-DATA-RULE-007	Closed bugs are not freely edited; Reopen is used when processing continues.
FRS-DATA-RULE-008	Comments do not directly change status.
FRS-DATA-RULE-009	History log must record important actions.

7. Acceptance Criteria Summary

Functional Area	Minimum acceptance criteria
Bug creation	Valid bug can be submitted as Assigned or Pending Assignment; invalid bug is blocked.
Duplicate support	User can search existing bugs before creation and decide follow, reopen, or create new.
Classification	Required classification fields and valid pair rules work.
Assignment	Developer list filters by responsibility and rejects invalid assignee.
Pending Assignment	Queue is visible to PM/Tester and can later be assigned.
Developer review	Developer can review, request information, reject, progress, and resolve.
Rejected follow-up	Rejected bug always has reason, nextProcessor, history, notification, and follow-up transition.
Retest and closure	Resolved bug can go to Retest Required; pass closes; fail reopens.
Comments	Comment stores author, role, timestamp, and content without status change.
History	Important actions show actor, time, old/new

Functional Area	Minimum acceptance criteria
Notifications	value, and reason where applicable. Event records exist without hardcoded external endpoints.
PM monitoring	PM can identify workload, overdue, pending, rejected, retest, and nextProcessor queues.
Advisory AI	Suggestions remain optional, review-only, safe on failure, and cannot mutate the bug.

8. Traceability to SRS

FRS ID	Related SRS IDs
FRS-BUG-001	SRS-FR-BUG-001, SRS-FR-BUG-002, SRS-DATA-001, SRS-DATA-002
FRS-BUG-002	SRS-FR-BUG-003, SRS-FR-BUG-004
FRS-CLASS-001	SRS-FR-CLASS-001 to SRS-FR-CLASS-005
FRS-ASSIGN-001	SRS-FR-ASSIGN-001, SRS-FR-ASSIGN-002
FRS-ASSIGN-002	SRS-FR-ASSIGN-003, SRS-FR-PM-003
FRS-ASSIGN-003	SRS-FR-ASSIGN-004, SRS-FR-STATUS-004
FRS-DEV-001	SRS-FR-STATUS-001, SRS-FR-STATUS-005
FRS-INFO-001	SRS-FR-STATUS-002, SRS-FR-STATUS-009
FRS-REJECT-001	SRS-FR-STATUS-003, SRS-FR-STATUS-004, SRS-FR-STATUS-009
FRS-STATUS-002	SRS-FR-STATUS-006, SRS-FR-STATUS-007
FRS-STATUS-003	SRS-FR-BUG-005, SRS-FR-STATUS-007
FRS-STATUS-005	SRS-FR-STATUS-008, SRS-NFR-INT-001
FRS-NEXTP-001	SRS-FR-STATUS-009
FRS-COMMENT-001	SRS-FR-COMMENT-001, SRS-FR-COMMENT-002
FRS-AUDIT-001	SRS-FR-AUDIT-001, SRS-FR-AUDIT-002
FRS-NOTIF-001	SRS-FR-NOTIF-001, SRS-FR-NOTIF-002
FRS-PM-001	SRS-FR-PM-001, SRS-FR-PM-002, SRS-FR-PM-003
FRS-PM-002	SRS-FR-PM-004
FRS-UX-001	SRS-IF-UI-001 to SRS-IF-UI-005, SRS-NFR-USE-001, SRS-NFR-USE-002

FRS ID	Related SRS IDs
FRS-AI-001	SRS-FR-AI-001 to SRS-FR-AI-003

9. Open Issues

ID	Open Issue	Owner	Functional Impact
OI-FRS-001	Confirm PM direct reassignment permission.	Team / Mentor	Determines whether PM sees Assign/Reassign actions or only request/comment actions.
OI-FRS-002	Confirm notification channel for MVP.	Team / Mentor	Determines whether notification records alone satisfy MVP.
OI-FRS-003	Confirm attachment storage approach.	Team / Mentor	Determines whether Attachments are metadata-only or actual file handling.
OI-FRS-004	Confirm overdue thresholds and workload limits.	Team / PM	Determines PM dashboard calculations.
OI-FRS-005	Closed: workflow diagrams are rendered as figures in the FRS and standalone Diagram Pack.	IDTS Project Team	Visual DOCX and Drive review pack are the current review baseline.